# 1 Project Proposal

## 1.1 Overview

Car Performance & Tuning Lab is a web-based simulation tool that allows car enthusiasts and mechanics to model the real-world impact of performance modifications before spending a single dollar. Users enter their vehicle's base specifications, select one or more modifications, and instantly see calculated performance deltas across key metrics — side by side, stock vs. modified.

## 1.2 Problem Statement

Currently, there is no reliable tool for predicting exactly how a specific modification will affect a particular car's performance profile. Enthusiasts rely on forums, word-of-mouth, and expensive trial-and-error. This project provides a data-driven, interactive alternative.

## 1.3 Objectives

- Build a real-time configuration panel that accepts vehicle specs (weight, HP, torque, drivetrain) withzero-latency feedback.
- Implement a back-end calculation engine (Node.js / Express) with calibrated coefficients for eachmodification type.
- Visualise performance changes through before/after bar charts, radar charts, and a downloadablesummary card.
- Support saving and comparing multiple builds within a single session.
- Deliver a fully bilingual (Arabic / English) UI with dark mode from day one.

## 1.4 Project Scope

| In Scope | Out of Scope |
|---|---|
| **Real-time performance calculation engine** | Real OBD-II / ECU hardware integration |
| **Turbo, Exhaust, ECU remap, Cold air intake mods** | Suspension, brake, or tyre modelling |
| **0–100 km/h time, top speed, fuel delta metrics** | Multi-vehicle comparisons across different makes |
| **Bar charts, radar chart, PDF export** | Mobile native apps (iOS / Android) |
| **Save / compare up to 5 builds** | User authentication / cloud storage |
| **Arabic & English localisation** | Paid / subscription features |

## 1.5 Stakeholders

| Stakeholder | Role / Interest |
|---|---|
| Ministry of Communications & IT | Programme sponsor — graduation criteria |
| Digital Egypt Initiative Supervisors | Evaluation and grading |
| Team228 (4 members) | Development and delivery |
| Car enthusiasts / end-users | Primary product users |

### 2.1 Gantt Chart (8-Week Timeline)

| Phase | Week | Key Activities | Deliverable | Owner |
|---|---|---|---|---|
| Research & Scoping | 1–2 | Define final scope, map component tree, set up GitHub repo & branching strategy, agree API contract format | Scope doc<br>Repo scaffold<br>API spec | All |
| UI Design | 2 | Figma wireframes for all screens, color system, typography tokens, responsive breakpoints defined | Figma file<br>Design system | Jomana<br>Jiovany |
| Backend Dev | 3 | Express server, TypeScript calculation engine, unit tests (Jest) for every formula | Working API (verified) | Youssef<br>Shady |
| Frontend Core | 4 | Config panel wired to live API, real-time recalculation, form validation, error states | Live panel | All |
| Charts & Viz | 5 | Bar charts (Recharts), radar chart (Chart.js), before/after comparison, summary card | Full UI | Jiovany<br>Jomana |
| Features | 6 | Save/compare builds, PDF export, Arabic/English i18n, dark mode toggle | Feature-complete | All |
| Polish & QA | 7 | Cross-browser testing, responsive QA, accessibility review, bug fixes, documentation | Production build | Shady<br>Youssef |
| Deploy | 8 | Frontend → Vercel, Backend → Render, demo video recorded, final report submitted | Live URL<br>Demo video | All |

### 2.2 Milestones

| # | Milestone | Target Date | Success Criteria |
|---|---|---|---|
| M1 | Project kick-off & repo live | End of Week 1 | All members can push/pull; scope agreed |
| M2 | API contract signed off | End of Week 2 | Figma + OpenAPI spec reviewed by superviso |
| M3 | Calculation engine passing all tests | End of Week 3 | Jest: 100% formula coverage, 0 failures |
| M4 | Config panel live with real API | End of Week 4 | Live recalculation demo recorded |
| M5 | All charts rendered correctly | End of Week 5 | All 3 chart types visible; data matches API |
| M6 | Feature freeze | End of Week 6 | i18n, PDF export, save/compare functional |
| M7 | QA sign-off | End of Week 7 | Zero P1 bugs; Lighthouse score ≥ 85 |
| # | Milestone | Target Date | Success Criteria |
| M8 | Deployment & final submission | End of Week 8 | Live URL accessible; demo video uploaded |

### 2.3 Resource Allocation

| Resource | Tool / Platform | Purpose | Cost |
|---|---|---|---|
| Version Control | GitHub (Free) | Source control, PRs, CI | Free |
| Frontend Hosting | Vercel (Hobby) | Production deployment | Free |
| Backend Hosting | Render (Free tier) | Express API hosting | Free |
| Design | Figma (Education) | Wireframes & design system | Free |
| Testing | Jest + Playwright | Unit + E2E tests | Free |
| Monitoring | Vercel Analytics | Page views, error tracking | Free |
| Communication | Discord + Notion | Daily standups, task tracking | Free |

### 3.1 Role Overview

| Member | Primary Role | Secondary Role |
|---|---|---|
| Youssef Malak | Backend Lead | DevOps / Deployment |
| Shady Kerolos | Frontend Lead | QA & Testing |
| Jiovany Isaac | Full-Stack Dev | Data Visualisation |
| Jomana Ahmed | UI/UX Designer | Localisation (i18n) |

### 3.2 Detailed Responsibilities

#### Youssef Malak — Backend Lead / DevOps

- Design and implement the Node.js + Express calculation engine in TypeScript.
- Write Jest unit tests for every performance formula (0-100, top speed, fuel delta).
- Define and document the REST API contract (OpenAPI / Swagger).
- Set up GitHub repository, branching strategy, and conventional commits policy.
- Deploy backend to Render; configure environment variables and CORS.
- Own the calculation coefficient calibration and accuracy review.

#### Shady Kerolos — Frontend Lead / QA

- Build the main React configuration panel with live state management.
- Wire all frontend components to the live calculation API.
- Implement form validation, loading states, and error boundaries.
- Lead cross-browser and responsive QA in Week 7.
- Write end-to-end tests using Playwright for critical user flows.
- Produce the final production build and Lighthouse audit report.

#### Jiovany Isaac — Full-Stack Developer / Data Viz

- Implement all chart components: bar charts (Recharts) and radar chart (Chart.js).
- Build the before/after comparison view and summary card component.
- Implement the PDF export feature for the summary card.
- Assist backend with API integration and debugging data shapes.
- Implement the save and compare multiple builds feature.
- Record the final demo video in Week 8.

#### Jomana Ahmed — UI/UX Designer / Localisation

- Create all Figma wireframes and the full design system (colors, typography, spacing).
- Define and implement the dark mode color tokens.
- Implement Arabic / English i18n using react-i18next.
- Ensure RTL layout correctness for Arabic mode.
- Own accessibility review (WCAG AA compliance) in Week 7.

- Produce all user-facing copy in both languages.

### 3.3 Collaboration Protocols

- Daily async standup in Discord: what did I do yesterday, what am I doing today, any blockers.
- All work is done in feature branches; no direct commits to main.
- Pull requests require at least one approval before merging.
- Conventional commits enforced: feat:, fix:, chore:, test:, docs:.
- Notion board tracks tasks in: Backlog / In Progress / In Review / Done.
- Weekly sync call every Sunday to review milestone status.

### 4.1 Risk Register

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| R1 | Calculation coefficients are inaccurate, producing unrealistic results | Technical | Medium | High | 9 | Validate every formula agains | |
| R2 | Real-time recalculation causes UI lag on low-end devices | Performance | Medium | Medium | 6 | Debounce API calls (300ms). | |
| R3 | Team member unavailable for 1+ we | Resourcek due to illness or exam | Medium | High | 9 | Cross-train members on each | |
| R4 | Render (free tier) cold-starts cause slow API response | Infrastructure | High | Medium | 8 | Add a keep-alive ping every 1 | |
| R5 | Chart.js radar chart has unexpected rendering issues | Technical | Low | Medium | 4 | Prototype radar chart in isolati | |
| R6 | PDF export library produces malformed output on some br | Technical | Medium | Low | 4 | Test jsPDF across Chrome, F | |
| R7 | RTL layout breaks in Arabic mode | UX | Medium | Medium | 6 | Use CSS logical properties (m | |
| R8 | Scope creep: requests for additional mod types or metrics | Scope | High | Medium | 8 | Scope is locked after Week 1. | |

### 4.2 Risk Scoring Matrix

Score = Likelihood × Impact, using: Low = 1, Medium = 2, High = 3. Scores 7–9: High priority — mitigate immediately. Scores 4–6: Medium priority — monitor weekly. Scores 1–3: Low priority — accept and log.

### 4.3 Contingency Plan

- **If calculation engine is not ready by end of Week 3:** Use hardcoded mock data for frontend development; backend catches up in parallel during Week 4.
- **If a team member drops out:** Redistribute tasks among remaining 3 members; inform supervisor immediately; scope may be reduced by removing PDF export.
- **If deployment fails in Week 8:** Use Netlify as Vercel fallback and Railway as Render fallback. Both are pre-configured in the repo.
- **If demo recording fails:** Loom screen record is always running during final testing as a backup.

## 5 Key Performance Indicators (KPIs)

### 5.1 Technical Performance KPIs

| KPI | Target | Measurement Method | Frequency |
|---|---|---|---|
| API Response Time (p95) | < 300ms | Render dashboard / browser devtools | Weekly from W3 |
| Frontend FCP | < 1.5s | Lighthouse CI on every PR | Per PR |
| Lighthouse Performance Score | ≥ 85 | Lighthouse audit | W7 + deploy |
| Lighthouse Accessibility | ≥ 90 | Lighthouse audit | W7 + deploy |
| System Uptime (30-day) | ≥ 99.5% | UptimeRobot free monitor | Continuous |
| Unit Test Coverage | 100% formulas | Jest coverage report | End of W3 |
| Zero P1 bugs at launch | 0 critical bugs | QA test log | End of W7 |

### 5.2 User Experience KPIs

| KPI | Target | Measurement Method |
|---|---|---|
| Recalculation latency (visible to user) | < 200ms after debounce | Manual stopwatch + Network tab |
| RTL layout correctness (Arabic mode) | 0 misaligned elements | Manual QA checklist |
| PDF export success rate | 100% across 3 browsers | Cross-browser test matrix |
| Dark/Light mode toggle works | No FOUC or flicker | Manual test + video recording |
| Build save/compare (up to 5 builds) | All builds persist in session | Manual functional test |
| Mobile responsiveness | Works on 375px–1440px | BrowserStack / DevTools |

### 5.3 Project Delivery KPIs

| KPI | Target | Notes |
|---|---|---|
| All 8 milestones hit on time | 100% | Tracked on Notion board |
| PR review turnaround | < 24 hours | GitHub PR logs |
| Conventional commit compliance | ≥ 95% of commits | GitHub commit history |
| Weekly sync attendance | 100% of members | Discord meeting log |
| Final submission on time | Week 8 deadline | Ministry submission portal |
| Demo video duration | 3–5 minutes | Covers all major features |

### 5.4 KPI Dashboard Summary

All technical KPIs will be tracked automatically via CI/CD pipelines. Lighthouse is run on every pull request via GitHub Actions. UptimeRobot sends alerts to the team Discord if uptime drops below threshold. Delivery KPIs are reviewed every Sunday in the weekly sync. Any KPI that misses its target triggers an

immediate root-cause discussion and corrective action logged in the project Notion page.

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